Final Exam A Solutions: Cumulative

GSE 510

Solutions

1. **Sets, Bounds, and Feasibility**

Consider

$$B = \{(x_1, x_2) \in \mathbb{R}^2_+ : 3x_1 + 2x_2 \leq 18\}.$$

(a) For $(2, 4)$,

$$3(2) + 2(4) = 6 + 8 = 14 \leq 18.$$

Also, both coordinates are nonnegative. Therefore,

$$(2, 4) \in B.$$

(b) For $(5, 2)$,

$$3(5) + 2(2) = 15 + 4 = 19 > 18.$$

Therefore,

$$(5, 2) \notin B.$$

(c) To find the largest possible value of x_1, set $x_2 = 0$. Then

$$3x_1 \leq 18,$$

so

$$x_1 \leq 6.$$

Therefore, the largest possible value of x_1 is

$$6.$$

(d) To find the largest possible value of x_2, set $x_1 = 0$. Then

$$2x_2 \leq 18,$$

so

$$x_2 \leq 9.$$

Therefore, the largest possible value of x_2 is

$$9.$$

(e) The set B is closed because it is defined using weak inequalities:

$$x_1 \geq 0, \qquad x_2 \geq 0, \qquad 3x_1 + 2x_2 \leq 18.$$

It is bounded because every feasible bundle satisfies

$$0 \leq x_1 \leq 6 \qquad \text{and} \qquad 0 \leq x_2 \leq 9.$$

Thus, B is closed and bounded.

2. **Logic and Sequences**

Consider the statement:

$$\text{If } x > 3, \text{ then } x^2 > 9.$$

(a) The converse is:

$$\text{If } x^2 > 9, \text{ then } x > 3.$$

(b) The contrapositive is:

$$\text{If } x^2 \leq 9, \text{ then } x \leq 3.$$

(c) The original statement is true. If $x > 3$, then x is positive and larger than 3, so

$$x^2 > 3^2 = 9.$$

(d) The converse is false. For example, if

$$x = -4,$$

then

$$x^2 = 16 > 9,$$

but

$$x = -4 \not> 3.$$

Therefore, $x^2 > 9$ does not imply $x > 3$.

(e) The sequence is

$$a_n = 3 + \frac{5}{n}.$$

Since

$$\frac{5}{n} \to 0$$

as $n \to \infty$, we have

$$\lim_{n\to\infty} a_n = 3.$$

Therefore, the sequence converges to

$$3.$$

3. **Open and Closed Sets**

Let

$$S = [1, 4).$$

(a) The set S is not open in $\mathbb{R}$. The point 1 belongs to S, but any open interval around 1 contains points less than 1, which are not in S.

(b) The set S is not closed in $\mathbb{R}$. It does not contain the boundary point 4.

(c) The boundary points of S are

$$1 \qquad \text{and} \qquad 4.$$

(d) One sequence in S that converges to a point not in S is

$$x_n = 4 - \frac{1}{n}.$$

For every n, $x_n \in [1, 4)$, and

$$\lim_{n\to\infty} x_n = 4.$$

But

$$4 \notin S.$$

(e) One sequence in S that converges to a point in S is

$$y_n = 2 + \frac{1}{n}.$$

For every n, $y_n \in [1, 4)$, and

$$\lim_{n\to\infty} y_n = 2.$$

Since

$$2 \in S,$$

the limit belongs to S.

4. **Linear Algebra and Systems**

Consider the system

$$2x_1 + 3x_2 = 16, \qquad x_1 + 4x_2 = 14.$$

(a) The system in matrix form is

$$\begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 16 \\ 14 \end{bmatrix}.$$

(b) The determinant is

$$|\mathbf{A}| = \begin{vmatrix} 2 & 3 \\ 1 & 4 \end{vmatrix} = 2(4) - 3(1) = 8 - 3 = 5.$$

(c) Since

$$|\mathbf{A}| = 5 \neq 0,$$

the matrix $\mathbf{A}$ is nonsingular.

(d) Solve the system:

$$2x_1 + 3x_2 = 16,$$
$$x_1 + 4x_2 = 14.$$

From the second equation,

$$x_1 = 14 - 4x_2.$$

Substitute into the first equation:

$$2(14 - 4x_2) + 3x_2 = 16.$$

Therefore,

$$28 - 8x_2 + 3x_2 = 16.$$

Thus,

$$28 - 5x_2 = 16,$$

so

$$-5x_2 = -12.$$

Hence,

$$x_2 = \frac{12}{5}.$$

Then

$$x_1 = 14 - 4\left(\frac{12}{5}\right) = \frac{70}{5} - \frac{48}{5} = \frac{22}{5}.$$

Therefore,

$$x_1 = \frac{22}{5}, \qquad x_2 = \frac{12}{5}.$$

(e) Nonsingularity means that the coefficient matrix has a nonzero determinant. For this system, it means the two equations are independent and there is a unique solution.

5. **Span and Linear Independence**

Let

$$\mathbf{v}_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \qquad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

(a) A general linear combination is

$$a\mathbf{v}_1 + b\mathbf{v}_2 = a \begin{bmatrix} 2 \\ 1 \end{bmatrix} + b \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

Therefore,

$$a\mathbf{v}_1 + b\mathbf{v}_2 = \begin{bmatrix} 2a + b \\ a + 3b \end{bmatrix}.$$

(b) The matrix with columns $\mathbf{v}_1$ and $\mathbf{v}_2$ is

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}.$$

Its determinant is

$$|A| = 2(3) - 1(1) = 6 - 1 = 5.$$

(c) Since

$$|A| = 5 \neq 0,$$

the vectors are linearly independent.

(d) Since there are two linearly independent vectors in $\mathbb{R}^2$, they span $\mathbb{R}^2$.

(e) We want

$$\begin{bmatrix} 2a + b \\ a + 3b \end{bmatrix} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}.$$

This gives the system

$$2a + b = 7,$$

$$a + 3b = 8.$$

From the first equation,

$$b = 7 - 2a.$$

Substitute into the second equation:

$$a + 3(7 - 2a) = 8.$$

Therefore,

$$a + 21 - 6a = 8.$$

Thus,

$$-5a = -13,$$

so

$$a = \frac{13}{5}.$$

Then

$$b = 7 - 2\left(\frac{13}{5}\right) = \frac{35}{5} - \frac{26}{5} = \frac{9}{5}.$$

Therefore, yes, the vectors can generate

$$\begin{bmatrix} 7 \\ 8 \end{bmatrix}$$

using

$$a = \frac{13}{5}, \qquad b = \frac{9}{5}.$$

6. **OLS in Matrix Form**

Suppose

$$\mathbf{y} = \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix}, \qquad \mathbf{X} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix}.$$

(a) First,

$$\mathbf{X}' = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}.$$

Therefore,

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 4 \\ 4 & 10 \end{bmatrix}.$$

(b)

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 6 \end{bmatrix} = \begin{bmatrix} 11 \\ 21 \end{bmatrix}.$$

(c) The determinant of $\mathbf{X}'\mathbf{X}$ is

$$3(10) - 4(4) = 30 - 16 = 14.$$

Therefore,

$$(\mathbf{X}'\mathbf{X})^{-1} = \frac{1}{14} \begin{bmatrix} 10 & -4 \\ -4 & 3 \end{bmatrix}.$$

(d) The OLS estimator is

$$\widehat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}.$$

Thus,

$$\widehat{\boldsymbol{\beta}} = \frac{1}{14} \begin{bmatrix} 10 & -4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix}.$$

Multiplying,

$$\begin{bmatrix} 10 & -4 \\ -4 & 3 \end{bmatrix} \begin{bmatrix} 11 \\ 21 \end{bmatrix} = \begin{bmatrix} 110 - 84 \\ -44 + 63 \end{bmatrix} = \begin{bmatrix} 26 \\ 19 \end{bmatrix}.$$

Therefore,

$$\widehat{\boldsymbol{\beta}} = \begin{bmatrix} 26/14 \\ 19/14 \end{bmatrix} = \begin{bmatrix} 13/7 \\ 19/14 \end{bmatrix}.$$

(e) The fitted regression line is

$$\widehat{y} = \frac{13}{7} + \frac{19}{14}x.$$

7. **Differentiation, Approximation, and Interpretation**

Suppose

$$C(q) = 100 + 12q + 2q^2.$$

(a) The marginal cost function is

$$C'(q) = 12 + 4q.$$

(b) At $q = 10$,

$$C'(10) = 12 + 4(10) = 52.$$

(c) A linear approximation gives

$$C(11) - C(10) \approx C'(10)(11 - 10).$$

Since

$$11 - 10 = 1,$$

we get

$$C(11) - C(10) \approx 52.$$

(d) Compute the exact values:

$$C(11) = 100 + 12(11) + 2(11)^2 = 100 + 132 + 242 = 474.$$

Also,

$$C(10) = 100 + 12(10) + 2(10)^2 = 100 + 120 + 200 = 420.$$

Therefore,

$$C(11) - C(10) = 474 - 420 = 54.$$

(e) The approximation is not exactly equal to the true change because the cost function is nonlinear. Marginal cost at $q = 10$ measures the instantaneous rate of change at $q = 10$, while the exact change from 10 to 11 depends on how marginal cost changes over that whole interval.

8. **Unconstrained Optimization with Two Variables**

Consider

$$f(x, y) = 40x + 30y - 2x^2 - y^2 - xy.$$

(a) The first partial derivatives are

$$f_x = 40 - 4x - y$$

and

$$f_y = 30 - 2y - x.$$

(b) The first-order conditions are

$$40 - 4x - y = 0$$

and

$$30 - 2y - x = 0.$$

Rearranging,

$$4x + y = 40$$

and

$$x + 2y = 30.$$

From the first equation,

$$y = 40 - 4x.$$

Substitute into the second equation:

$$x + 2(40 - 4x) = 30.$$

Thus,

$$x + 80 - 8x = 30.$$

Therefore,

$$-7x = -50,$$

so

$$x = \frac{50}{7}.$$

Then

$$y = 40 - 4\left(\frac{50}{7}\right) = \frac{280}{7} - \frac{200}{7} = \frac{80}{7}.$$

The critical point is

$$\left(\frac{50}{7}, \frac{80}{7}\right).$$

(c) The second partial derivatives are

$$f_{xx} = -4, \qquad f_{yy} = -2, \qquad f_{xy} = -1.$$

Therefore, the Hessian matrix is

$$H = \begin{bmatrix} -4 & -1 \\ -1 & -2 \end{bmatrix}.$$

(d) The determinant is

$$|H| = (-4)(-2) - (-1)(-1) = 8 - 1 = 7.$$

Since

$$|H| > 0$$

and

$$f_{xx} = -4 < 0,$$

the critical point is a local maximum.

(e) The value at the critical point is

$$f\left(\frac{50}{7}, \frac{80}{7}\right).$$

Substituting,

$$f = 40\left(\frac{50}{7}\right) + 30\left(\frac{80}{7}\right) - 2\left(\frac{50}{7}\right)^2 - \left(\frac{80}{7}\right)^2 - \left(\frac{50}{7}\right)\left(\frac{80}{7}\right).$$

This simplifies to

$$f\left(\frac{50}{7}, \frac{80}{7}\right) = \frac{2600}{7}.$$

9. **Differential Equation and Stability**

Consider

$$\frac{dp}{dt} = 0.4(D(p) - S(p)),$$

where

$$D(p) = 90 - 2p, \qquad S(p) = 30 + p.$$

(a) Substitute demand and supply into the differential equation:

$$\frac{dp}{dt} = 0.4\left[(90 - 2p) - (30 + p)\right].$$

(b) Simplify:

$$(90 - 2p) - (30 + p) = 60 - 3p.$$

Therefore,

$$\frac{dp}{dt} = 0.4(60 - 3p).$$

Hence,

$$\frac{dp}{dt} = 24 - 1.2p.$$

(c) The steady-state price satisfies

$$24 - 1.2p^* = 0.$$

Therefore,

$$1.2p^* = 24,$$

so

$$p^* = 20.$$

(d) The steady state is stable. If

$$p < 20,$$

then

$$24 - 1.2p > 0,$$

so price rises. If

$$p > 20,$$

then

$$24 - 1.2p < 0,$$

so price falls. Therefore, price moves toward 20.

(e) Economically, price rises when demand exceeds supply and falls when supply exceeds demand. The steady state is the market-clearing price where demand equals supply.

10. **Solow-Style Difference Equation**

Consider

$$k_{t+1} = sAk_t^{1/2} + (1-\delta)k_t,$$

where

$$s = 0.5, \qquad A = 8, \qquad \delta = 0.25.$$

(a) Substitute the parameter values:

$$k_{t+1} = 0.5(8)k_t^{1/2} + (1-0.25)k_t.$$

Therefore,

$$k_{t+1} = 4k_t^{1/2} + 0.75k_t.$$

(b) A steady state satisfies

$$k^* = 4(k^*)^{1/2} + 0.75k^*.$$

Rearranging,

$$k^* - 0.75k^* = 4(k^*)^{1/2}.$$

Thus,

$$0.25k^* = 4(k^*)^{1/2}.$$

If $k^* > 0$, divide both sides by $(k^*)^{1/2}$:

$$0.25(k^*)^{1/2} = 4.$$

Therefore,

$$(k^*)^{1/2} = 16.$$

Squaring both sides,

$$k^* = 256.$$

There is also the zero steady state:

$$k^* = 0.$$

The positive steady state is

$$k^* = 256.$$

(c) Suppose $k_0 = 64$. Then

$$k_1 = 4\sqrt{64} + 0.75(64).$$

Since

$$\sqrt{64} = 8,$$

we get

$$k_1 = 4(8) + 48 = 32 + 48 = 80.$$

(d) Suppose $k_0 = 400$. Then

$$k_1 = 4\sqrt{400} + 0.75(400).$$

Since

$$\sqrt{400} = 20,$$

we get

$$k_1 = 4(20) + 300 = 80 + 300 = 380.$$

(e) Capital moves toward the positive steady state $k^* = 256$. If

$$k_0 = 64 < 256,$$

then

$$k_1 = 80 > 64,$$

so capital rises toward the steady state. If

$$k_0 = 400 > 256,$$

then

$$k_1 = 380 < 400,$$

so capital falls toward the steady state.

(f) Economically, the positive steady state is the level of capital where investment exactly offsets depreciation, so the capital stock remains constant from one period to the next.